

Experimental design handover

Cluster-randomised trial · 48 clinics · SPIRIT protocol · design rationale, sample size with design effect, randomisation plan and ethics package

This is a redacted sample. It shows the structure and level of detail of a real deliverable. Client, setting and clinical details have been removed or changed. Ask us and we will send one built around a study like yours.

1 Design rationale

The document researchers do not know to ask for and reviewers always want. It records every design considered and why it was rejected, dated before recruitment — so that when a reviewer asks why you did not run a stepped wedge, the answer already exists in your own documentation.

Design	Verdict, and why
Individually randomised trial	Not possible. The intervention is a change to how a clinic triages, so it reaches every patient in that clinic. This single fact selected the unit of randomisation before anything else was considered.
Stepped-wedge cluster trial	Considered seriously, because every clinic would eventually receive the intervention and the trust would have found it easier to agree to. Rejected because the rollout would have run across a winter and the outcome is strongly seasonal — time and treatment would have been confounded.
Cluster crossover	Rejected. A clinic cannot un-learn a triage protocol; the effect does not reverse, so the washout a crossover depends on does not exist.
Before-and-after at each clinic	Rejected. No concurrent control, so any change in referral patterns over the study year would be indistinguishable from the intervention.
Parallel cluster-randomised trial	Chosen. Concurrent control, no confounding with season, and a unit of randomisation that matches how the intervention is delivered.

What the design gives up, stated in the protocol. Statistical efficiency. Forty-eight clinics carry considerably less information than 896 independently randomised patients would, which is why the sample is almost sixty per cent larger. Saying so in the protocol is better than leaving a reviewer to point it out.

2 Sample size

Element	Value and source
Primary outcome	Proportion triaged correctly within four hours, at twelve months
Control rate	64%, from a twelve-month audit across nine of the participating clinics
Target difference	10 percentage points, agreed with the clinical group as the smallest worth changing practice for
Power and alpha	90%, two-sided 5%

Element	Value and source
If randomised individually	448 per arm
Design effect	$1 + (30 - 1) \times 0.02 = 1.58$, ICC taken from two published cluster trials
Inflated for clustering	708 per arm
Clinics required	24 per arm
Clinics recruited	50 in total, allowing for two withdrawing entirely
Sensitivity	Recomputed at ICC 0.04, which would require 33 clinics per arm. Stated in the protocol as the principal risk to power.

The target difference is the number that decides everything else, and it is the one most often chosen backwards from what is affordable. Here it was agreed with the clinical group before the calculation was run, and the agreement is recorded.

3 Threats to validity

Separating what the design solves from what it can only declare. A protocol that addresses the first and admits the second is considerably more convincing than one that is silent about both.

Closed by design

- **Identification bias.** Patients are identified and consented before their clinic is told its allocation. A clinic that knows its arm can otherwise select who enters the study, and in cluster trials this is the most common fatal flaw.
- **Unblinded outcome assessment.** Triage appropriateness is judged from the record by assessors who are not told the clinic's arm. The clinicians cannot be blinded; the assessors can, and that is the blinding that protects the primary outcome.
- **Chance imbalance.** With 24 clusters in each arm, imbalance on clinic size and baseline performance remains likely. Allocation is stratified on both and the analysis adjusts for them.

Declared, not solved

- **Contamination.** Staff move between clinics within the trust, so control clinics may partially adopt the protocol. This biases towards the null and cannot be designed out; it is stated as a limitation and measured through a staff movement log.
- **External validity.** All 48 clinics sit in one health system with one records platform. The protocol says so, rather than implying the result will transfer to systems that work differently.

Agreed with the ethics committee in advance. Cluster allocation means individual patients cannot consent to their arm. The consent model — clinic-level opt-in with individual consent for data collection — was settled with the committee before submission, which is what avoided a second round.

4 What is handed over

- The design rationale, recording every design considered and the reason each was or was not chosen.
- The full protocol, written to SPIRIT with the CONSORT cluster extension in view.
- The sample size calculation, its sources, and the sensitivity analysis around the ICC.
- The randomisation plan: stratified allocation, sequence generation and the concealment procedure.

- The statistical analysis plan, with the mixed model, covariates, missing data strategy and pre-specified subgroups.
- The ethics package: committee submission, consent documents and participant information sheets.
- The trial registry entry, drafted and ready to file before the first participant is enrolled.

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